PAPER ID-411411

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BTECH
(SEM IV) THEORY EXAMINATION 2023-24
THEORY OF AUTOMATA AND FORMAL LANGUAGES

TIME: 3 HRS **M.MARKS: 70**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

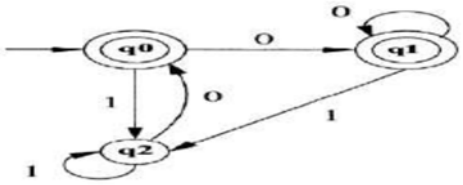
SECTION A

1. Attempt *all* questions in brief. 2 x 7 = 14

a.	Give the mathematical definition of DFA. Differentiate between NFA and DFA.
b.	Construct Deterministic Finite Automata (DFA) to accept string that always ends with 101 over alphabet Σ ={0,1}
c.	Give regular expressions that represent the language (L), which has all binary strings having two consecutive 0s and two consecutive 1s over the alphabet Σ = {0, 1}.
d.	Compute the Language generated by the given CFG G = ({S}, {a, b}, P, S} where P is defined by: {S → SS, S → ab, S → ba, S → ε}
e.	Let G be the grammar S → 0B | 1A A → 0 | 0S | 1AA B → 1 | 1S | 0BB Determine the leftmost derivation for the string 00110101
f.	Explain the concept of two stack PDA. Give an example of a language that is accepted by two stack PDA but not accepted by normal one stack PDA.
g.	Explain Multi Tape Turing Machine.

SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

a.	Construct a Finite automata (DFA) which accepts all binary numbers whose decimal equivalent is divisible by 4 over Σ = {0, 1}.
b.	Compute the regular expression using Arden`s Theorem for the following DFA. 
c.	Write an equivalent left linear grammar from the given right linear grammar. S→0A |1B A→0C |1A |0 B→1B |1A |1 C→0 |0A
d.	Differentiate between DPDA and NPDA. Construct a PDA that accepts language L = $\{a^n b^n \mid n \geq 1\}$.
e.	Differentiate between Deterministic Turing machine and Non-Deterministic Turing machine. Design a Turing machine for the language L={ww | w ε (a + b)*}.

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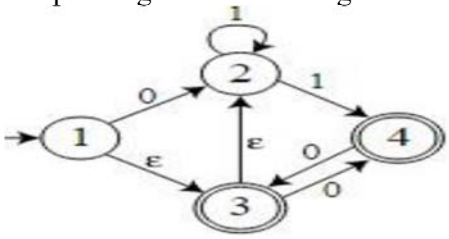
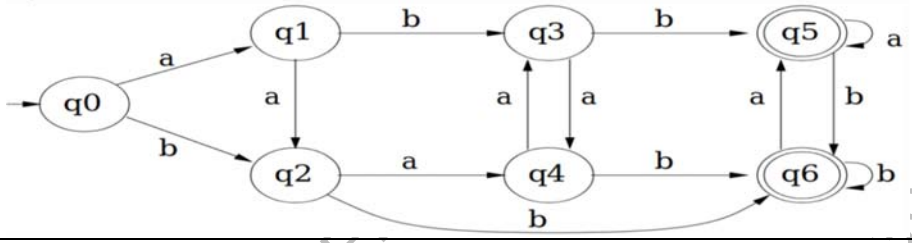
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SECTION C

3. Attempt any *one* part of the following: **7 x 1 = 7**

a.	Construct a DFA corresponding to the following NFA with ϵ moves: 
b.	Express in the minimum state automata equivalent to DFA described in below figure: 

4. Attempt any *one* part of the following: **7 x 1 = 7**

a.	State Pumping Lemma for Regular Language. Show that the given language $L=\{a^p \mid$ Where p is a prime$\}$ is not regular.
b.	Discuss closure properties (i.e. union, concatenation, complement, intersection and difference) of regular language.

5. Attempt any *one* part of the following: **7 x 1 = 7**

a.	Reduce the given grammar G = ({S, A, B}, {a, b}, P, S) to Chomsky Normal form. Where P is defined by: S →bA | aB A → bAA | aS | a B →aBB | bS | b
b.	Design a CFG for the following language: **(i)** $L=\{0^m 1^n \mid m \neq n$ & m, n>=1$\}$ **(ii)** $L=\{a^p b^q c^r \mid p+q=r$ & p, q > = 1$\}$

6. Attempt any *one* part of the following: **7 x 1 = 7**

a.	Construct PDA equivalent to the following CFG G = ({S, A}, {0,1}, P, S} where P is defined by: S →0S1 | A A →1A0 | S | ε

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b.	Find the equivalent CFG of the following PDA P = ({q0, q1,}, {a, b}, {a, z0}, δ, q0, z0) where δ is given by: δ (q0, a, z0) = (q0, az0) δ (q0, a, a) = (q1, aa) δ (q1, a, a) = (q1, ε) δ (q1, ε, z0) = (q1, ε)

7. Attempt any *one* part of the following: **7 x 1 = 7**

a.	Construct Turing Machine that accepts language L=$\{a^{2n}b^n \mid n>=1\}$. Also show the instantaneous description for the string w = aaaabb.
b.	Explain the any two of the following: i. Universal Turing Machine. ii. Post Correspondence Problem. iii. Recursive and recursively Enumerable Languages

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